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UGC - Autonomous Institute
Accredited by NBA & NAAC with 'A' Grade
Approved by AICTE
Permanently affiliated to JNTUH

PROJECT BASED LEARNING

ON

CIRCULAR WATER TANK DESIGN BY USING AUTOCAD SOFTWARE

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APPROVED BY AICTE, AFFILIATED TO JNTUH, ACCREDITED BY NBA & NAAC WITH A-GRADE)

MAISAMMAGUDA(V), KOMPALLY, SECUNDERABAD, 500100. (2026-2027)

ABSTRACT

Storage tanks are critical infrastructure for distribution systems, ensuring a reliable water supply for domestic, industrial, and agricultural needs. Among various configurations, circular water tanks are widely preferred due to their high structural efficiency and uniform stress distribution. This paper presents the analysis and design of a reinforced concrete circular water tank resting on the ground. The structural evaluation focuses on resisting hoop tension and bending moments caused by internal hydrostatic pressure. Using analytical methods and compliance with standard design codes, the thickness of the tank wall, base slab dimensions, and reinforcement detailing are determined. The study compares fixed and flexible base connections to evaluate their impact on structural performance and cost-effectiveness. The findings demonstrate that circular geometry minimizes material consumption while optimizing load distribution. Ultimately, this project establishes a safe, durable, and economical design framework, ensuring long-term structural integrity and minimal maintenance for public water storage solutions.

INTRODUCTION

Circular water tanks are specialized storage structures designed to hold water for domestic, industrial, and agricultural use. They are among the most common geometric shapes used in civil engineering for fluid storage due to their structural efficiency.

The primary advantage of a circular design is its ability to resist internal hydrostatic pressure uniformly. Unlike rectangular tanks, which experience high stress concentrations at the corners, circular tanks distribute hoop tension evenly across the walls. This uniform distribution minimizes the risk of cracking and structural failure, making them highly durable.

These tanks can be constructed using various materials, including reinforced concrete, steel, and polyethylene. Depending on the space availability and system requirements, they can be built ground-supported, underground, or elevated. Their optimized shape requires less material for construction compared to rectangular tanks of the same capacity, making them a cost-effective and reliable solution for modern water management.

AIM & OBJECTIVES

AIM :

To analyze, design, and construct a structurally sound and leak-proof circular water tank that efficiently stores and distributes water while optimizing material usage and space.

OBJECTIVES :

- **Optimize Structural Integrity:** Design a stable structure to withstand hydrostatic pressure, soil pressure, and environmental loads.
- **Ensure Water-Tightness:** Select high-grade concrete mixes and appropriate waterproofing agents to prevent any leakage or seepage.
- **Maximize Storage Efficiency:** Utilize a circular geometry to achieve the highest volume-to-surface-area ratio, minimizing material costs.
- **Analyze Stress Distribution:** Evaluate hoop tension and bending moments to determine the exact reinforcement requirements for the tank walls and base.
- **Promote Longevity:** Implement durable, corrosion-resistant materials to ensure low maintenance and a long operational lifespan.
- **Maintain Water Quality:** Integrate proper ventilation and drainage systems to prevent contamination and facilitate routine cleaning.

DRAWING DIMENSIONS-1

1. Dimension Breakdown Vertical Dimensions (Heights & Thicknesses)PCC Thickness: 100 mm Footing Thickness: 450 mm Wall Height (Below Ground Level): 2000 mm Wall Height (Above Ground Level): 1000 mm Total Tank Internal Height: 3000 mm (\(2000 + 1000\))Top Slab Thickness: 150 mm Horizontal Dimensions (Diameters & Widths)PCC Base Width (Square): 4200 mm Footing Width (Square): 4000 mm Tank Outer Diameter: 2500 mm (\(\ varnothing 2500\))Tank Inner Diameter: 2040 mm (\(\ varnothing 2040\))RCC Wall Thickness: 230 mm [calculated as \((2500 - 2040) / 2\)]Manhole Diameter: 600 mm (\(\ varnothing 600\)

DRAWING DIMENSION-2

2. AutoCAD Commands Guide To Draw the Plan View (Bottom Drawing)CIRCLE (C):Draw the inner wall: Specify center point \(\right arrow \) type D (Diameter) \(\right arrow \) enter 2040.Draw the outer wall: From the same center \(\right arrow \) type D \(\right arrow \) enter 2500.Draw the manhole: Offset from the center \(\right arrow \) type D \(\right arrow \) enter 600.RECTANGLE (REC):Draw the footing base: Specify start point \(\right arrow \) type D (Dimensions) \(\right arrow \) enter 4000 for length and 4000 for width .Draw the PCC base: Specify start point \(\right arrow \) type D \(\right arrow \) enter 4200 for length and 4200 for width. 2. AutoCAD Commands Guide To Draw the Plan View (Bottom Drawing)CIRCLE (C):Draw the inner wall: Specify center point \(\right arrow \) type D (Diameter) \(\right arrow \)

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DRAWING DIMENSION-3

To Draw the Section View (Top Drawing)LINE (L) / POLYLINE (PL): To draw the horizontal baselines for PCC, footing, and ground level (G.L).OFFSET (O):Offset PCC baseline upward by 100 for PCC thickness .Offset footing baseline upward by 450 for footing thickness .Offset lines inward by 230 to create the RCC side wall thickness .Offset top wall line upward by 150 for the top slab .TRIM (TR): To clean up overlapping lines at the intersections of the walls, footing, and slab .HATCH (H): Use concrete hatch patterns (AR-CONC) to represent the RCC side walls, footing, slab, and PCC layers .DIMLINEAR (DLI) / DIMDIAMETER (DDI): To apply the linear measurements and diameter annotations exactly as shown /.MTEXT (T): To add labels like "600 MANHOLE", "G.L", and "RCC SIDE WALLS".

TECHNOLOGY AND USES

The technology

Circular water tanks utilize **cylindrical geometry** to uniformly distribute hydrostatic pressure across the tank walls. This eliminates localized stress points or weak corners. The wall behaves primarily under **hoop tension**, reducing structural bulging and the risk of cracks. [1, 2, 3, 4, 5]

Engineering materials include **Reinforced Cement Concrete (RCC)**, **High-Density Polyethylene (HDPE) plastics**, and **corrugated galvanized steel**. Their optimized geometry features a minimal circumference-to-volume ratio. This lowers material requirements and improves cost-efficiency. Furthermore, the lack of sharp corners prevents dirt accumulation and microbe growth, ensuring simple hydraulic maintenance.

DRAWING TOOLS

- LINE
- CIRCLE
- RECTANGLE

- TEXT
- MTEXT
- DIMENSIONS
- HATCH

  LAYER TOOLS
- LAYER PROPERTIES
- LINE TYPE
- LINE WEIGHT
- COLOUR MANAGEMENT

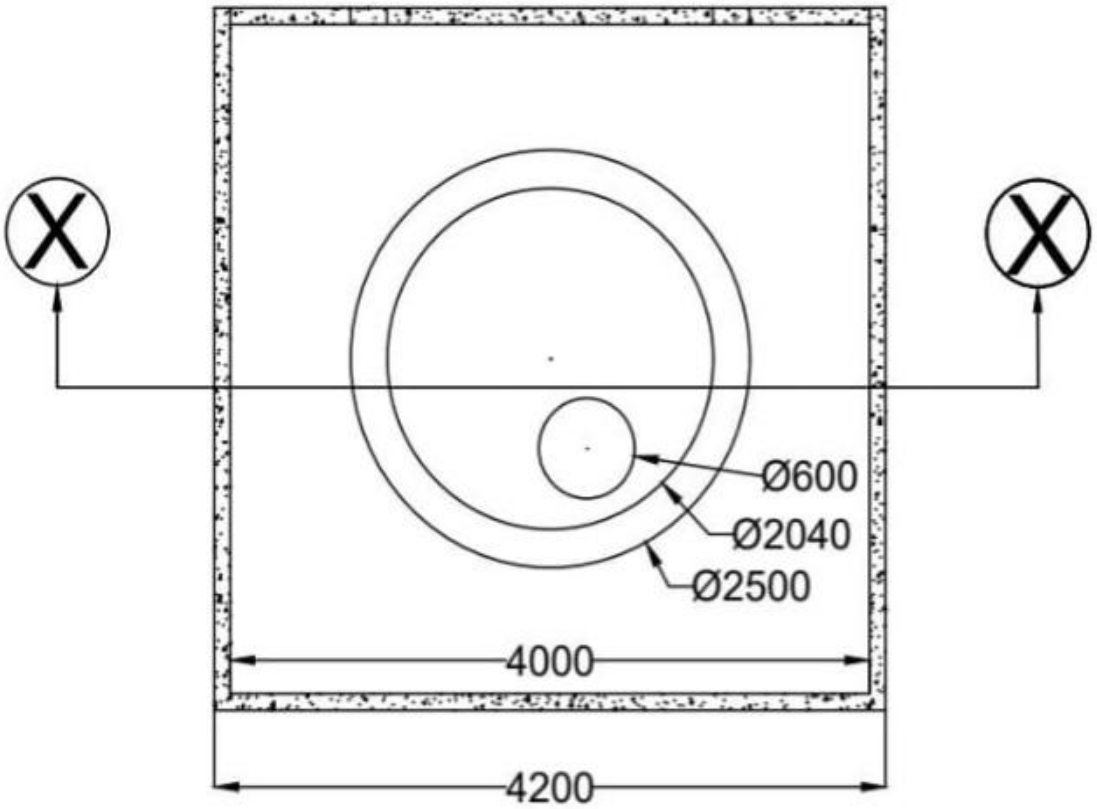
MODIFY TOOLS

- OFFSET
- TRIM
- MIRROR
- SCALE
- EXTEND
- SCALE
- STRECH
- MOVE
- COPY

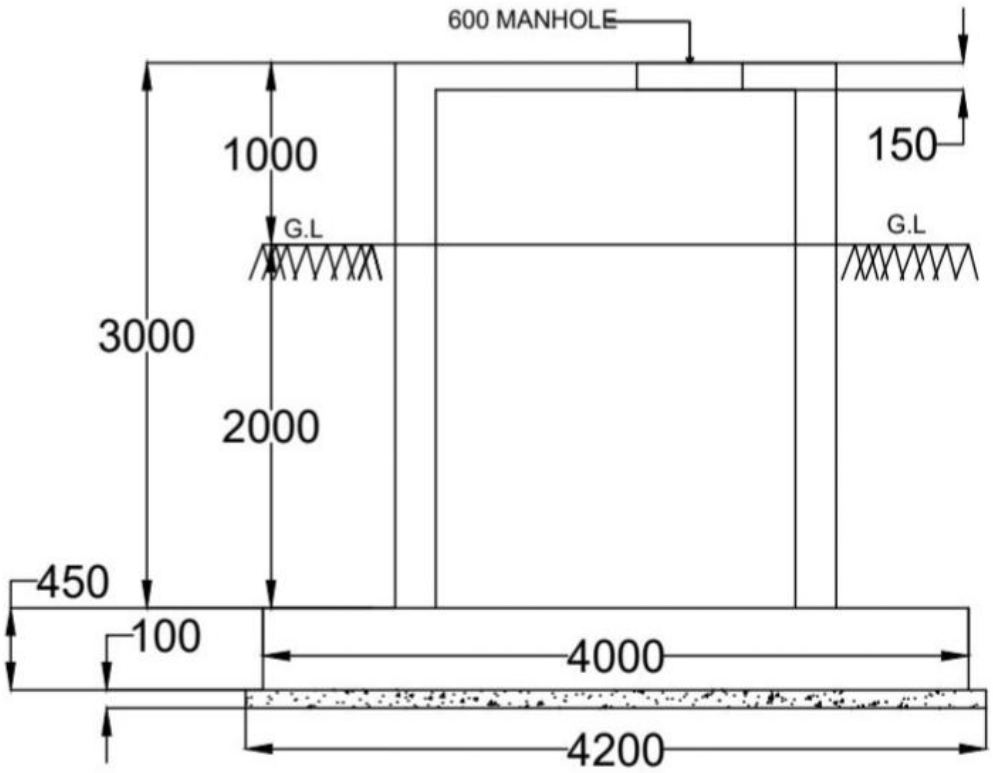
APPLICATIONS

AutoCAD helps engineers design precise circular water tanks for civil and industrial projects. Key applications include :Structural detailing: Creating accurate 2D floor plans, elevations, and rebar layouts.3D modelling : Visualizing tank geometry, wall thickness, and dome roofs .Site integration: Placing tanks into water treatment plant layouts and piping networks .Estimation: Extracting dimensions for volume and material calculations.

CIRCULAR WATER TANK

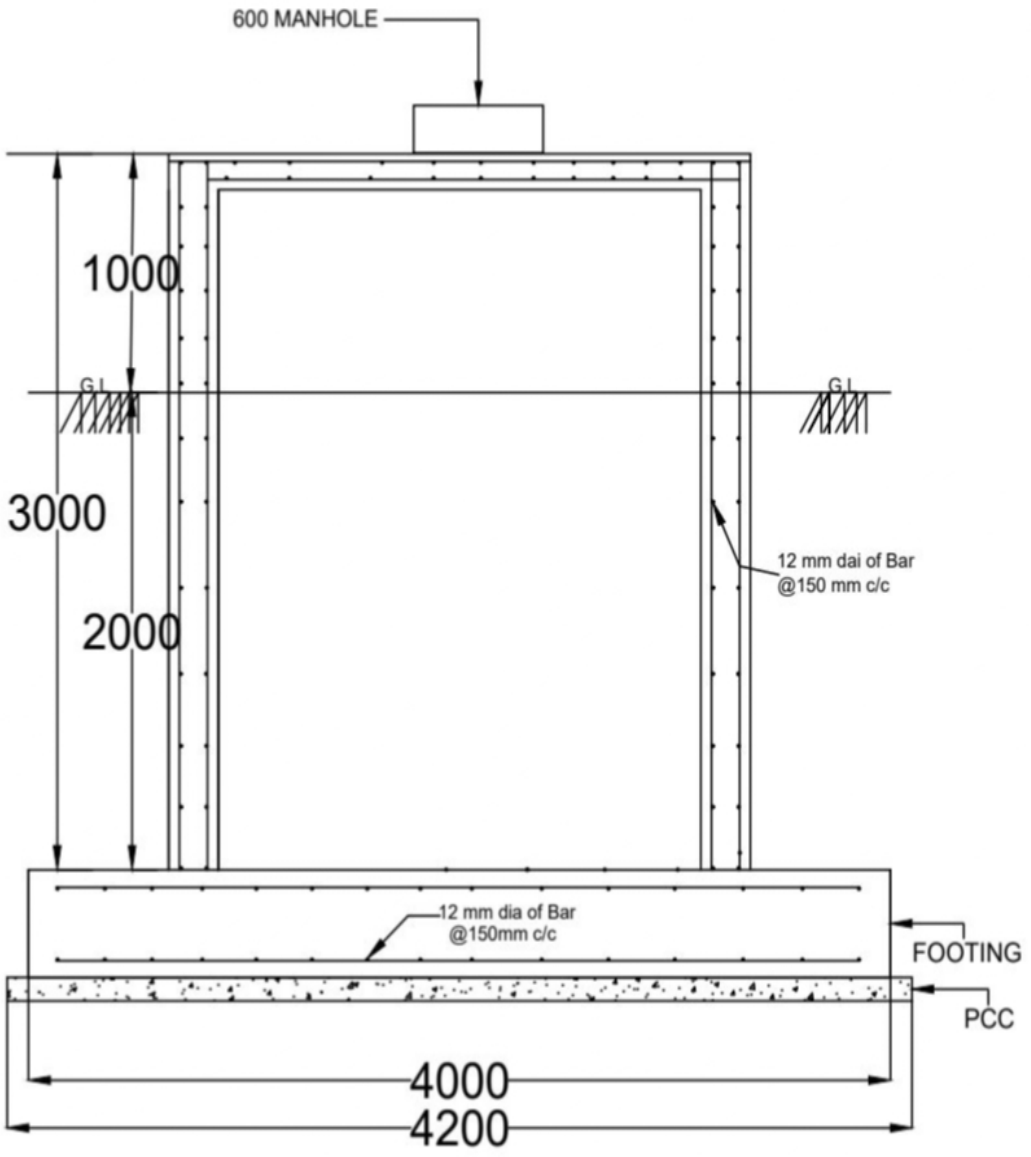


PLAN



SECTION X - X`

CIRCULAR WATER TANK - RCC DETAILING



ADVANTAGES

Perfect symmetry: Easily generated using a single center point and radius .Efficient modification: Uniform scaling and offsetting streamline wall thickness adjustments . Precise calculations: The AREA command instantly extracts accurate volume data . Simplified detailing: Polar array replicates structural reinforcement bars effortlessly . Fewer coordinates: Reduces complex drafting time compared to rectangular tanks.

RESULT

Plan View: Displays outer diameter, inner diameter, and wall thickness . Elevation View: Shows tank height, water level, and footing details .Details: Includes reinforcement layout, inlet/outlet pipe positions, and dome or flat roof specifications ready for structural analysis and site execution.

CONCLUSION

This project successfully finalized the structural and architectural design of a circular water tank using AutoCAD. The software ensured high drafting precision, allowing for accurate scaling of the cylindrical wall, concrete base, and dome roof. AutoCAD's parametric constraints and detailing tools optimized reinforcement placement to effectively resist hydrostatic pressure and hoop tension. The generated 2D engineering drawings and 3D models provide an error-free blueprint ready for construction. Ultimately, utilizing AutoCAD streamlined the design workflow, reduced manual drafting errors, and ensured compliance with structural safety standards for reliable fluid storage.